

JAVA SOFTWARE ENGINEER BOOTCAMP

6-Week Immersive Enterprise Java Full-Stack Program





INSTRUCTOR: DR. GURINDERJEET KAUR

- 2015–Present: 11+ years of experience in academia, research, industry, and corporate training
- 2015–2023: University Professor teaching Computer Science and AI-related disciplines
- 2019–2022: PhD specialization in AI, Machine Learning, Deep Learning, and GPU Computing
- 2023: Machine Learning Researcher at Sony Research India
- 2024: Machine Learning Engineer for a US-based telecommunications startup
- 2024–Present: Corporate Trainer in AI/ML, Generative AI, MLOps, Cloud, and Cybersecurity
- 2025–July 2026: Artificial Intelligence Instructor at CIMT College, Canada
- Present: Contract-based Technical Trainer delivering enterprise training programs globally

COURSE ABSTRACT

- This six-week immersive program prepares early-career professionals for enterprise Java full-stack developer roles. The program builds foundational and practical skills across Java, Spring Boot, Apache Kafka, React, PostgreSQL, OpenShift, and CI/CD automation. The focus is hands-on and job-ready, equipping participants to contribute effectively to Java delivery teams from day one.
- Throughout the program, learners progress from Java fundamentals and object-oriented design through enterprise backend development with Spring Boot, event-driven architecture with Kafka, resilience patterns, full-stack integration with React and PostgreSQL, and cloud-native delivery using containers, Kubernetes, and OpenShift. Automated testing with JUnit, Mockito, and Selenium, CI/CD with Bitbucket Pipelines and Code Insights, infrastructure automation with Ansible and Terraform, and security testing are integrated throughout to reflect real enterprise delivery pipelines.
- Professional skills are woven throughout, including technical communication, stakeholder collaboration, pair programming, feedback practices, and developer accountability. The bootcamp culminates in a team-based capstone project where participants design, build, secure, automate, and deploy a full-stack Java application using the complete enterprise toolchain.

AUDIENCE

- Early-career professionals entering a Java full-stack developer role. Participants will develop, test, and deploy software solutions; design software to address business needs; prepare technical and procedural documentation; and maintain and debug software in an enterprise environment.

PREREQUISITES AND EXPECTATIONS

Prerequisites:

- Fundamental programming concepts (variables, control flow, functions, and basic data structures)
- Basic database concepts (tables, rows, and queries)

Expectations:

- Attend all sessions punctually and stay engaged for the full class window
- Come prepared: complete pre-lab exercises before each lab session
- Keep your camera on and actively participate in discussions and pair work
- Submit lab deliverables and checkpoints on time
- Communicate blockers early — ask questions in class or via the course channel
- Maintain a professional, collaborative, and respectful learning environment

COURSE WEEKS AGENDA

6-Week Program Structure:

- WEEK 1 — JAVA AND JVM FOUNDATIONS
- WEEK 2 — BACKEND, AI TOOLS AND TESTING
- WEEK 3 — SPRING FRAMEWORK AND ENTERPRISE PATTERNS
- WEEK 4 — KAFKA, REACT, POSTGRESQL AND RESILIENCE
- WEEK 5 — DEVOPS, CI/CD AND OPENSIFT
- WEEK 6 — CAPSTONE PROJECT

LEARNING OBJECTIVES

- Build backend services using Java and Spring Boot
- Develop and consume SOAP web services using standard enterprise patterns
- Implement event-driven systems using Apache Kafka and Spring Boot
- Develop React frontends integrated with Java APIs
- Persist data using PostgreSQL with Spring Data JPA
- Apply automated testing with JUnit and Mockito
- Implement CI/CD pipelines with Bitbucket Pipelines and Code Insights
- Automate infrastructure with Ansible and Terraform
- Deploy containerized applications to Kubernetes and OpenShift
- Apply application security testing practices
- Demonstrate professional communication, collaboration, and ownership skills
- Apply resilience and fault-tolerance patterns to enterprise services
- Deliver a team-based full-stack capstone project

LOGISTICS

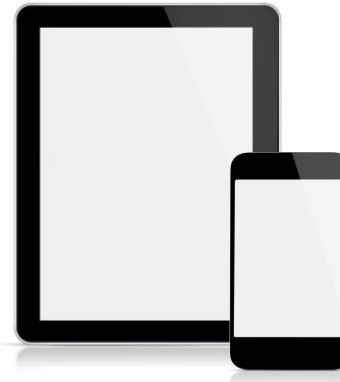
Class Hours:

- Class starts at 10:00 AM EST and finishes by 6:00 PM EST
- There will be regular breaks throughout the day



Telecommunication:

- Turn off or set electronic devices to silent (not vibrate)
- Reading or attending to devices can be distracting to other students
- Try to delay checking devices until breaks or after class



LET'S INTRODUCE OURSELVES

**Going around one by one,
please share:**

- Your name and current role or background
- Your experience with Java or programming so far
- One thing you're hoping to learn or achieve in this bootcamp
- A fun fact about yourself

